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BRIEF REPORT

Brief Report: Use of Superheroes Social Skills to Promote Accurate Social Skill Use in Children with Autism Spectrum Disorder

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Abstract The current study evaluated the use of Superheroes Social Skills to promote accurate use of discrete social skills in training and generalization conditions in two children with autism spectrum disorder. Participants attended a twice weekly social skills training group over 5 weeks, with lessons targeting nonverbal, requesting, responding, and conversation skills. A multiple probe across social skills design, replicated across participants, was utilized to assess the effects of participation of the intervention on skill accuracy. Following introduction of the intervention, participants demonstrated abrupt improvements in skill accuracy in both training and generalization conditions. Additionally, parental reports of participant social functioning indicated improvements following participation in the intervention. Limitations and future directions are discussed.

Keywords Social skills training · Video modeling · Self-monitoring · Superheroes Social Skills

Introduction

Autism spectrum disorder (ASD) is characterized by impairments in social communication, and restricted repetitive behaviors and interests (American Psychiatric Association 2013). Impairments in social skill development result in poor outcomes in children and adults with ASD, such as poor academic achievement, isolation, mental health problems, and unemployment (e.g., Bauminger et al. 2003; Howlin et al. 2000; Strain and Schwartz 2001). In order to improve outcomes for individuals with ASD, social skills training programs are frequently implemented with children with ASD (e.g., Goin-Kochel et al. 2007). Although many training programs successfully promote discrete social skill acquisition (e.g., Doggett et al. 2013), research suggests social skills training is often ineffective in promoting use of target social skills in generalized contexts (Bellini et al. 2007; Rao et al. 2008).

Superheroes Social Skills (Jenson et al. 2011) is a social intervention that trains for generalization through incorporation of generalization technologies described by Stokes and Osnes (1989). In an examination of the intervention, Radley et al. (2014b) evaluated the effect of participation in the Superheroes Social Skills program on accurate demonstration of discrete social skills in a training and a generalization setting with three children with ASD. Probes of accuracy in the training setting indicated immediate increases in level and trend of skill accuracy for all participants and skills. Generalization probes were also conducted, in which novel researchers presented novel cues for skill use and provided performance feedback following inaccurate demonstration of the target social skills. Generalization data also indicated immediate increases in level and trend of skill accuracy across all participants and skills. Probes of skill maintenance indicated that skills were maintained at levels similar to intervention.

Although results of Radley et al. (2014b) indicated improvements in discrete skill accuracy in both training and generalization probes, probes of generalization were dissimilar to real-world social skill use, as generalization was assessed by researchers who provided performance

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feedback. The purpose of the present study was to replicate findings and address limitations of Radley et al. by evaluating the effect of participation in the intervention on skill accuracy within a training setting, and generalization of accurate skill use to communication partners with whom patterns of communication were previously established who did not provide performance feedback.

Methods

Participants and Setting

This study was approved by the affiliate university's Institutional Review Board. Participants included two children with diagnoses of ASD. At the time of the study, Taylor, an 11-year-old male, had diagnoses of ASD and ADHD from a licensed psychologist. A review of records revealed that Taylor obtained a full scale IQ of 110 on the Wechsler Intelligence Scale for Children-Fourth Edition (Wechsler 2003). On the Kaufman Test of Educational Achievement-Second Edition (Kaufman and Kaufman 2004), he obtained standard scores of 117 on Reading and 120 on Math. A parent-completed Asperger Syndrome Diagnostic Scale (Myles et al. 2001) revealed an Asperger Syndrome Quotient of 115, suggesting a high probability of Asperger's Syndrome. Results from an occupational therapy evaluation revealed average to above average scores in visual motor integration and visual perception, while a speech-language evaluation revealed oral language skills in the average range. At the time of the study, Taylor received special education and gifted services at school, and had previously received clinic-based services for disruptive home behavior and impulse control. At the time of the study, Robby, a 12-year-old male, had diagnoses of ASD and moderate intellectual disability from a licensed psychologist. A review of records revealed that Robby obtained a full scale IQ of 60 on the Stanford Binet-Fifth Edition (Roid 2003). On the Wechsler Individual Achievement Test-Third Edition (Wechsler 2009), Robby obtained a Total Achievement standard score of 61. A parent completed Gilliam Autism Rating Scale-Second Edition (Gilliam 2006) revealed an overall Autism Index score of 124, suggesting a high probability of Autism. Additionally, Robby received a Global Adaptive Composite of 48 on a parent-completed Adaptive Behavior Assessment System-Second Edition (Harrison and Oakland 2003). He received special education services at school, and had previously received clinic-based services for disruptive home behaviors and daily living skills training.

Social skills training took place in a university-based clinic in the southeastern United States. Both participants were referred to the clinic for social skills training. All social skills training and probes of skill accuracy were conducted in a clinic-based conference room. A table and chairs were placed at the center of the room, parallel to a television that was used to display instructional videos. Generalization of skill acquisition was measured across persons and settings in the clinic hallway, a clinic treatment room, and in a clinician's office.

Dependent Measures

Probes of Skill Accuracy

Skill accuracy in the training setting was the primary dependent variable in the present study. During each probe of skill accuracy, a trained graduate student engaged the participant with a cue for a specific target social skill (e.g., "Taylor, what did you do at school today?") while another trained graduate student coded accurate demonstration of skill steps. The skill steps were coded using task analyses of target skills derived from the Superheroes Social Skills manual (Table 1), and the percentage of correctly demonstrated skill steps was calculated. During the intervention phase, two probes were typically delivered per participant per social skills training session, with one additional probe being delivered as necessary to demonstrate data stability.

Generalized skill accuracy was assessed across settings and individuals. Probes of generalized skill accuracy were delivered by non-researcher communication partners, consisting of parents and clinic personnel (e.g., administrative assistant) with whom the participant typically interacted upon entering the clinic. Generalization probes took place in a clinic waiting room or adjoining hallway using novel cues for skill use (e.g., "Robby, what is your favorite movie?). During generalized skill probes, the communication partner delivered a cue for target skill use while a trained graduate student coded for accurate demonstration of skill steps.

Social Functioning

Prior to and immediately following intervention, parents of participants completed the Autism Social Skills Profile (ASSP; Bellini and Hopf 2007). The ASSP consists of 49 4-point Likert items and results in a Total Social Functioning score comprised of three sub-scores: Social Reciprocity; Detrimental Social Behaviors; and Participation/Avoidance. Previous evaluations of the ASSP have found the measure to have high internal consistency ($\alpha = 0.94$) and test–retest reliability of 0.90 (Bellini and Hopf).

Table 1 Task analyses of target social skills

Body Basics	Expressing Wants and Needs	Responding to Questions and Requests	Conversation Maintenance
1. Face the person with head and shoulders oriented towards conversation partner	1. Initiate Body Basics steps	1. Stop current activity	1. Initiate Body Basics steps upon cue from conversation partner (e.g., "I like to watch movies"
2. Make and sustain eye contact for a minimum of 3 s	2. Signal to conversation partner that you want to talk (e.g., raise hand, tap on shoulder, say name)	2. Acknowledge question by initiation of Body Basics steps	2. Make an appropriate comment or ask a question about what the person says (e.g., "What is your favorite movie?"
3. Use an appropriate voice with volume appropriate for setting	3. Wait for the person to respond to the signal	3. Maintain eye contact while the other person is speaking	3. Maintain eye contact while the other person is speaking
4. Use the right expressing by having facial expression match conversation	4. Say what you want or need (e.g., "I need…"	4. Provide an appropriate response to the question	4. Make an appropriate comment or ask a question about what the person says (e.g., "I like that move too, it is really funny")
5. Relax, evidenced by relaxed shoulders, even paced speech, and not playing with objects	5. Wait for the person to respond		
	6. Acknowledge their reply (e.g., okay, thank you)		

Procedures

Design

A multiple probe across skills design with concurrent replication across participants (Gast 2010) was used. Phase changes were made once both participants reached mastery of the target skill in intervention, indicated by three consecutive probes of 100 % accuracy in the training setting.

Baseline

Prior to collection of baseline data, parents completed the ASSP independently. Analysis of parent-completed ASSPs indicated salient deficits in eye contact, asking questions, maintaining give and take of conversations, talking about interests of others, and requesting assistance. Salient deficits identified by parents of both participants were then matched with social skills included in the Superheroes Social Skills program. A total of four skills from the Superheroes Social Skills program were selected to be taught due to correspondence with parent-identified deficits on the ASSP: Body Basics (nonverbal skills), Expressing Wants and Needs, Responding to Questions and Requests, and Conversation Maintenance. Following identification of skills to be taught, baseline skill accuracy probes were conducted in both the training and generalization settings using task analyses of skills to be targeted using the Superheroes Social Skills program. Participants were not provided with praise or performance feedback during baseline. Phase change from baseline to intervention was made contingent upon both participant's data trend and variability.

Intervention

Skills were trained and probed during 1.5 h sessions over 5 weeks, with intervention sessions occurring twice weekly. Social skills training was presented in a group format, such that both participants received simultaneous training in the same target skill.

During the first lesson of each week, participants received didactic instruction in the target skill steps by animated superheroes via DVD. Animated didactic instruction consisted of an animator's hand holding a marker while rapidly drawing superhero characters, with voiceover narration and text providing rationale for and instruction in target skill use (Fig. 1). The participants then viewed two to three video models of unknown, similarly-aged peers performing the target skill in multiple contexts. Video models contained voiceover narration and text providing instruction for use of the skill (Fig. 1). Following, the facilitators reviewed skill steps and modeled correct and incorrect skill usage, encouraging participants to identify mistakes or omissions in skill steps when target skills were incorrectly modeled. Participants were then provided with self-monitoring cards to be marked contingent upon correct demonstration of the target skill steps. Next, participants role-played the skills with each other and the trainers, with praise and performance feedback provided. During role-plays, participants were encouraged to utilize the self-monitoring card upon accurate demonstration of the target skill. Role-play ended contingent upon two consecutive role-plays in which participants demonstrated 100 % skill accuracy. Participants then played a social game that required use of the target skill (e.g., Go Fish for Expressing Wants and Needs). Lastly, participants watched an

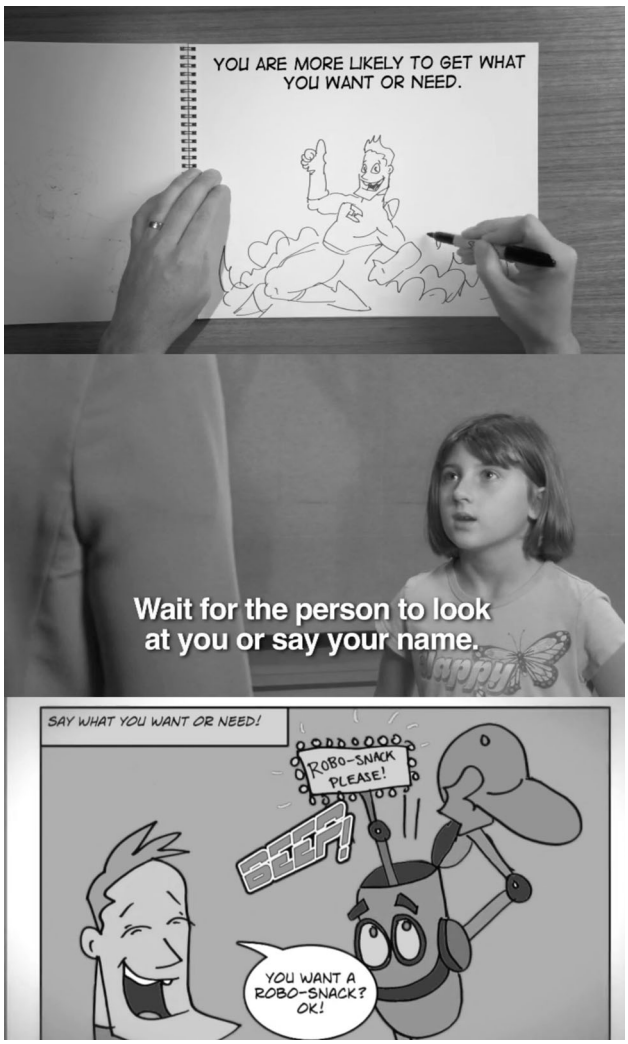


Fig. 1 Example of animated superhero providing skill rationale, a video model, and an animated comic for Expressing Wants and Needs skill

animated comic that reviewed the target skill (Fig. 1). Social skills training during the second session of each week was identical to the first, except that the social game was not played. Probes of skill accuracy in the training and generalization setting were completed at the end of each session.

Maintenance

Following demonstration of skill mastery, previously taught skills were probed in both settings to determine the extent to which the skill steps were maintained over 4 weeks for the Body Basics skill, 3 weeks for the Wants and Needs skill, and 2 weeks for the Responding skill, with length of maintenance phase varying across skills due to use of a multiple probe design. Procedures for maintenance phase probes were identical to baseline probes.

Interobserver Agreement

Interobserver agreement (IOA) was obtained by trained graduate students independently and simultaneously with the primary observer. IOA was calculated by dividing the number of agreements of skills demonstrated by the number of agreements plus disagreements and multiplied by 100. IOA was calculated during 42 % of observations for Taylor and 58 % of observations for Robby. Overall IOA for Taylor was 97 % (range 50–100 %) and 100 % for Robby. In addition, Kappa was also calculated and was found to be at 0.94 for Taylor (95 % CI 0.89–0.98) and 1.00 for Robby (95 % CI 1.00–1.00).

Treatment Integrity

Treatment integrity was assessed during each intervention session via an intervention manual-derived treatment integrity checklist. The checklist was completed by the primary facilitator and another trained graduate student independently throughout each session. Intervention procedures were found to be conducted with 100 % integrity for each session. Additionally, IOA for treatment integrity was 100 %.

Data Analysis

The effects of participation in the intervention on target social skill accuracy were evaluated via visual analysis of level, trend, and variability of data.

Nonoverlap of all pairs (NAP; Parker and Vannest 2009), a nonparametric method for determining the degree of data overlap between each intervention point in a phase (e.g., baseline) and each intervention point in a comparison phase (e.g., intervention), was also calculated using procedures described by Parker and Vannest. NAP scores between 0.00 and 0.65 are considered weak, from 0.66 to 0.92 are moderate, and scores of 0.93–1.00 are considered strong effects.

Results

Skill Acquisition

Taylor's skill accuracy data across each skill were variable during baseline (Fig. 2). Implementation of the Superheroes Social Skills program resulted in immediate and abrupt improvements in level and reductions in variability in training and generalization probes. During the maintenance phase, Taylor demonstrated maintained improvements in skill accuracy in both training and generalization

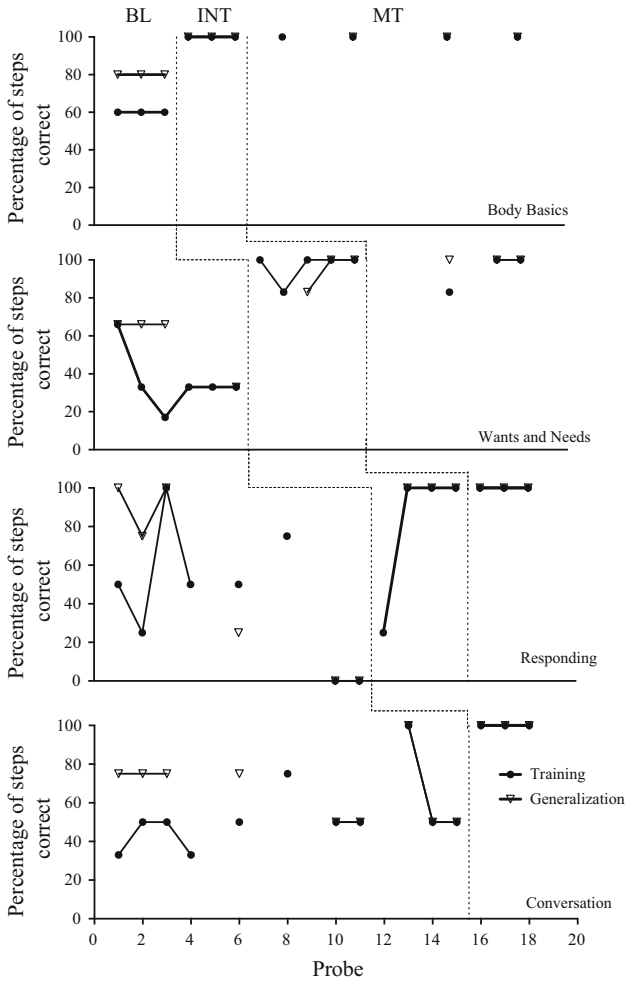


Fig. 2 Percentage of steps correctly demonstrated in training and generalization settings, Taylor

settings. Calculation of NAP revealed improvements ranging from moderate to strong across each skill and setting (Table 2).

Robby demonstrated low and variable levels of skill accuracy during baseline (Fig. 3). Immediate and abrupt improvements in accuracy were noted following implementation of intervention in both training and generalization probes. During maintenance phases, Robby demonstrated maintained improvements in skill accuracy for all skills probed. Calculation of NAP indicated strong intervention effects across skills and settings (Table 2).

Social Functioning

Social functioning of participants was assessed via the ASSP, administered to parents prior to and immediately following intervention. Both participants demonstrated clinically significant improvements on the ASSP Total Score and Reciprocity subscale, with moderate improvements on the Participation/Avoidance subscale (Table 3). Whereas Taylor's score demonstrated moderate improvements on the Detrimental Social Behaviors subscale, Robby's score demonstrated minimal improvement following intervention.

Discussion

The present study sought to replicate and address the limitations in assessment of generalization of Radley et al. (2014b). Both participants in the study demonstrated immediate increases in level and trend of skill acquisition in the training setting following introduction of intervention. In probes of generalization to communication partners with whom patterns of communication were previously established, both participants demonstrated abrupt improvements in skill accuracy without the provision of performance feedback. This finding expands upon results of Radley et al., which indicated improvements in

Table 2 NAP effect size for skill acquisition

	Taylor		Robby	
	Treatment	Generalization	Treatment	Generalization
Body Basics intervention	**1.00** (0.16–1.84)	**1.00** (0.16–1.84)	**0.97** (0.33–1.53)	**1.00** (0.23–1.78)
Body Basics maintenance	**1.00** (0.23–1.78)	**1.00** (0.16–1.84)	**1.00** (0.29–1.71)	**1.00** (0.23–1.78)
Expressing Wants and Needs intervention	**1.00** (0.39–1.60)	**1.00** (0.23–1.78)	**1.00** (0.26–1.64)	**1.00** (0.26–1.74)
Expressing Wants and Needs maintenance	**1.00** (0.29–1.71)	**1.00** (0.23–1.78)	**1.00** (0.29–1.71)	**1.00** (0.26–1.74)
Responding to Questions intervention	0.78 (0.04–1.17)	0.83 (−0.04 to 1.38)	**0.93** (0.28–1.36)	**1.00** (0.26–1.74)
Responding to Questions maintenance	**0.94** (0.20–1.55)	0.83 (−0.04 to 1.38)	**1.00** (0.31–1.69)	**1.00** (0.26–1.74)
Conversation intervention	**0.95** (0.27–1.55)	**0.94** (0.23–1.55)	**0.97** (0.39–1.48)	**1.00** (0.33–1.67)

NAP effect size scores below 0.65 are considered weak, scores between 0.66 and 0.92 are considered moderate, and scores ranging from 0.93 to 1.00 are considered strong (Parker and Vannest 2009). Strong effects in bold. Ninety percent confidence intervals in parentheses

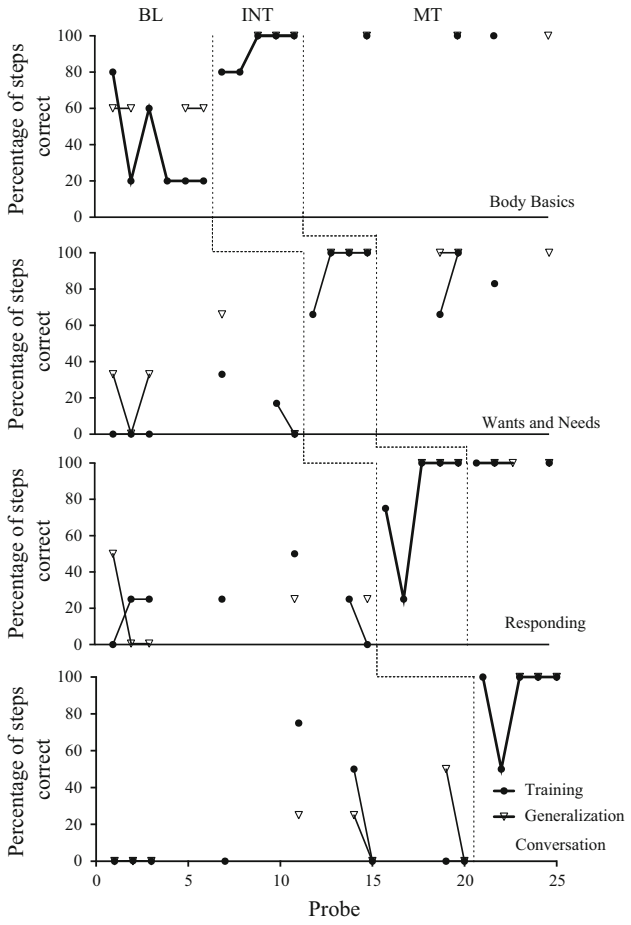


Fig. 3 Percentage of steps correctly demonstrated in training and generalization settings, Robby

Table 3 Pre- and post-intervention ASSP scores

Scale	Taylor		Robby	
	Pre	Post	Pre	Post
Total Social Functioning	119	142	92	108
Social Reciprocity	58	69	37	47
Participation/Avoidance	25	29	25	28
Detrimental Social Behaviors	25	30	23	24

generalized skill use in the presence of performance feedback. Additionally, accurate skill use was maintained in both settings. However, maintenance data in the current study were collected over a short period and conclusions regarding maintenance are therefore limited. Demonstration of generalization and short-term maintenance are, however, notable, as prior research suggested that generalization of skills is one of the most challenging aspects of social skills training (Barry et al. 2003) and social skills training has limited impact on generalization and maintenance of skills (Bellini et al. 2007). Generalization observed in the current study could be due to Superheroes Social Skills' incorporation of generalization technologies identified by Stokes and Osnes (1989), namely: the use of current functional contingencies (i.e. use of natural consequences of social skills use), training diversely (i.e. social game, video models), and engaging functional mediators of behavior (i.e. self-monitoring).

Both participants demonstrated improvements in post-intervention Total Social Functioning scores on the parent-completed ASSP, which suggests that Taylor and Robby may have utilized the targeted social skills outside of the clinical setting (i.e. home, community). Similar to data collected regarding improvements in skill accuracy following social skills training, improvements in social functioning noted by parents on the ASSP support previous research indicating generalized skill use following participation in the intervention (e.g., Radley et al. 2014a, b). Overall, data collected in the current study indicate that participation in the Superheroes Social Skills intervention resulted in improved social competence for both participants. It is, however, important to note that parent raters were not blind to treatment conditions and outcomes should be interpreted with this in mind.

Although results of the study indicate improved accuracy and generalization of target social skills, results should be viewed in light of limitations. First, the present study included two participants. Due to the small sample size, conclusions regarding the generalizability of these findings are limited. Future research should seek to evaluate the effects of participation in the intervention with a larger group of participants. Next, the Superheroes Social Skills program involves multiple components to support acquisition and generalization of social skills. As such, it is difficult to identify which of the components was functionally related to behavioral change. Further investigation of the Superheroes Social Skills program strategies via a component analysis is warranted. The four target social skills were taught in the same order for both participants. It is possible that some skills may be prerequisite for learning other skills, and changing the sequence of skills may result in different outcomes. Therefore, future studies may use counterbalancing treatment designs to address this limitation. Despite improvements in skill accuracy observed during generalization probes, the fact that all generalization probes took place within the clinic setting should be considered. As attendance to the social skills group may have served as a discriminative stimulus for skill use, future research should collect generalization probes in settings that are clearly different than training settings (e.g., home). Because ASSP data were collected as a pre- post- measure with no control group, external variables may have contributed to the observed improvements in parental report of

social functioning and data should be interpreted with caution. Additionally, as parents were not blind to treatment, rater expectation may have biased results. Finally, long-term follow-up data were not collected as part of the study. Long-term follow-up data of acquired skills would assist in making conclusions about maintenance and add to the generalizability of the results. Future research may address this limitation by collecting long-term maintenance data.

In summary, the intervention utilized in the current study resulted in improvements in discrete social skill accuracy, for two children, in a training setting. More notable, improvements in skill accuracy were observed during probes delivered by communication partners with whom patterns of communication were previously established and who did not provide performance feedback. These findings are divergent from those that have suggested social skills training to be minimally effective in promoting generalized and maintained behavior change (e.g., Bellini et al. 2007), and results may be, in part, attributed to the incorporation of multiple generalization technologies (Stokes and Osnes 1989) in the Superheroes Social Skills program. As programming for generalization appears to have been beneficial for the two participants in the current study, researchers and practitioners working with individuals with ASD are encouraged to consider incorporation of similar strategies to promote skill generalization.

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